Highlights

Field accuracy is not heat accuracy: wall-flux-constrained surrogates for non-Boussinesq convection

- Field error below 1% can coexist with $> 2\%$ Nu error in 43% of cavity samples.
- Wall-flux loss reduces matched six-point error against a non-converged 64^2 target.
- In a two-seed 4-by-4 matrix, every trained stencil is optimal only at its own order.
- A 134-family bootstrap leaves magnitude and temperature-curvature sign unresolved.
- Contrast scaling and 3D tests are consistency checks, not general validation.